

WebUp Industrial Engineering - Documentation

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1. Introduction

Welcome to the WebUp Industrial Engineering documentation. This project is a modern, accessible, and SEO-optimized static website showcasing metal fabrication and industrial engineering projects.

1.1. Project Overview

WebUp is a professional HTML5/CSS3 website designed for industrial engineering companies specializing in metal fabrication, CNC machining, and manufacturing services.

Key Features

- Fully responsive design with mobile ($\leq 480\text{px}$) and tablet ($\leq 768\text{px}$) breakpoints
- Fixed navigation bar with button-style menu and dropdown
- Active page indicators in navigation
- Accessibility compliant (WCAG standards)
- SEO optimized with meta tags and sitemap
- Privacy compliant (robots.txt, noindex tags)
- Modern CSS with variable-based theming
- Interactive image gallery with paper shadow effects
- Scrolling footer with W3C validator badges

1.2. Technology Stack

Technology	Description
HTML5	Semantic markup with ARIA labels
CSS3	Modern styling with CSS variables and media queries
Responsive Design	3 breakpoints: Desktop ($>768\text{px}$), Tablet ($\leq 768\text{px}$), Mobile ($\leq 480\text{px}$)
Git	Version control on branch v0.5.1
PlantUML	Architecture diagram generation
PlantUML	Architecture diagram generation



For detailed responsive layout information with visual examples, see [Responsive Design Guide](#).

2. Getting Started

2.1. Prerequisites

- Web browser (Chrome, Firefox, Safari, Edge)
- Text editor (VS Code recommended)
- Basic knowledge of HTML/CSS

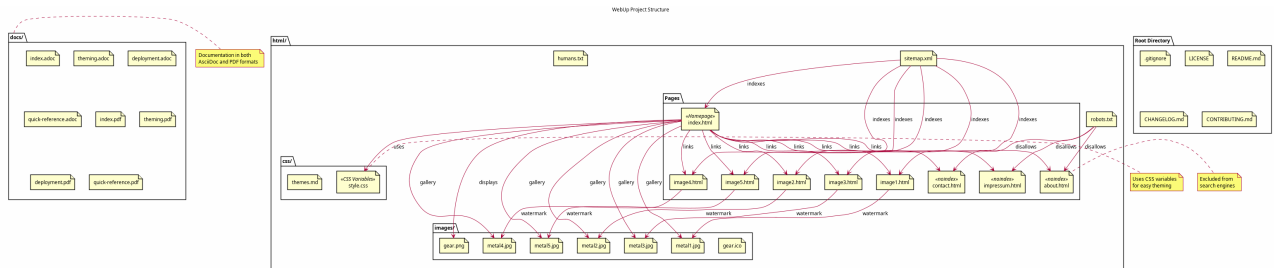
2.2. Installation

- ## 1. Clone the repository:

```
cd webup
```

2. Switch to the v0.5.1 branch:

```
git checkout v0.5.1
```



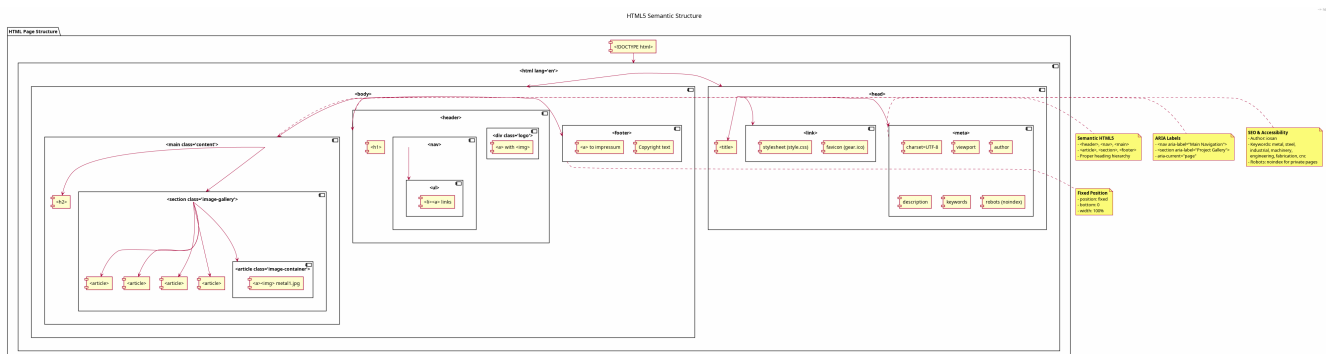
```
cd html
python -m http.server 8000
```

1. Navigate to <http://localhost:8000>

2.3. Quick Start

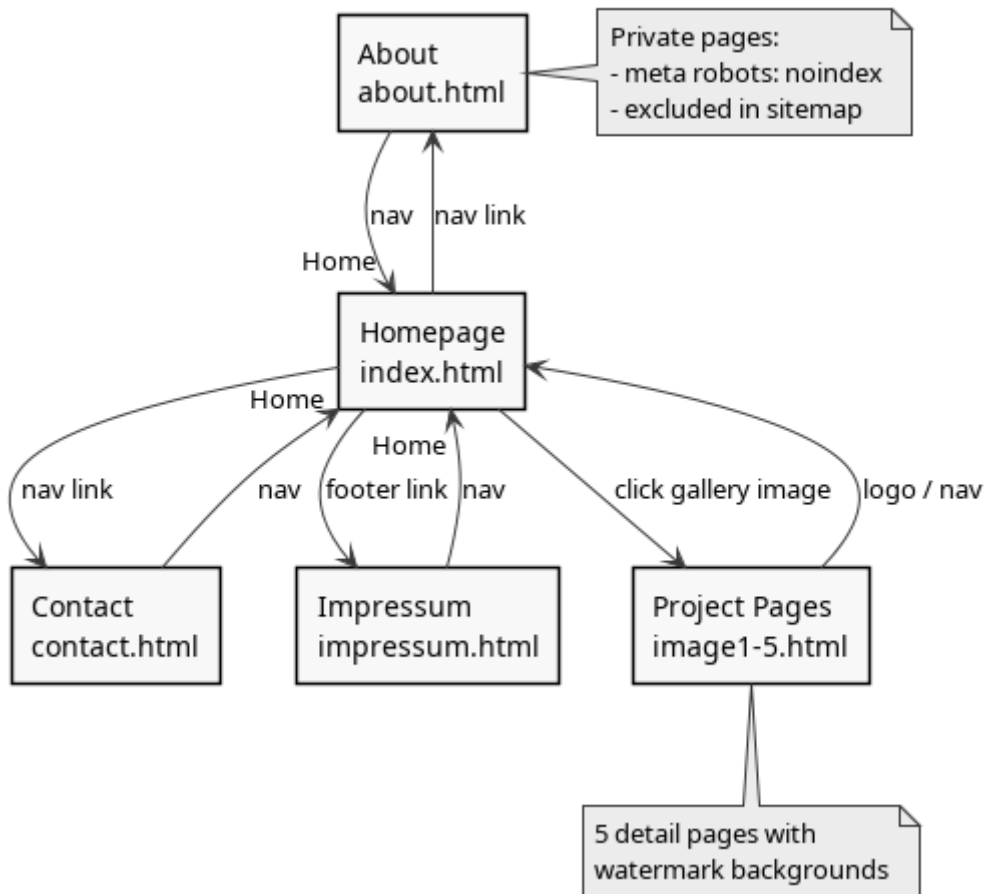
Open any HTML file in the `html/` directory directly in your browser. No build process required!

3. Project Structure



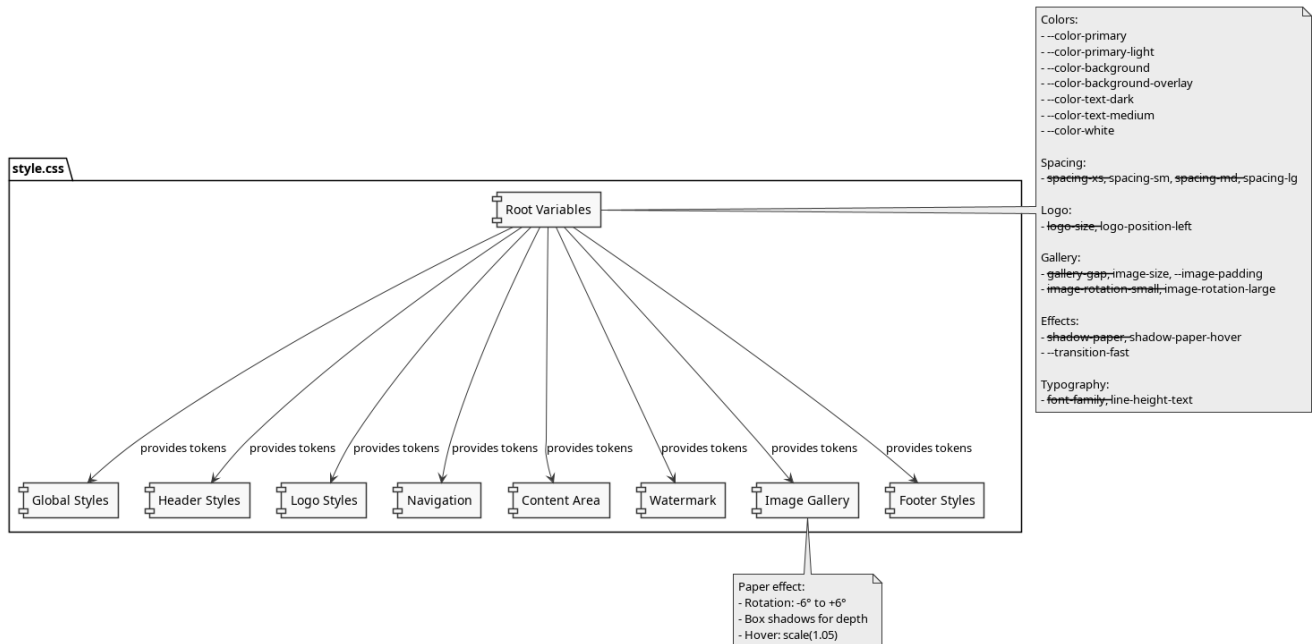
3.1. Navigation Flow

WebUp Navigation Flow (Simplified)



```

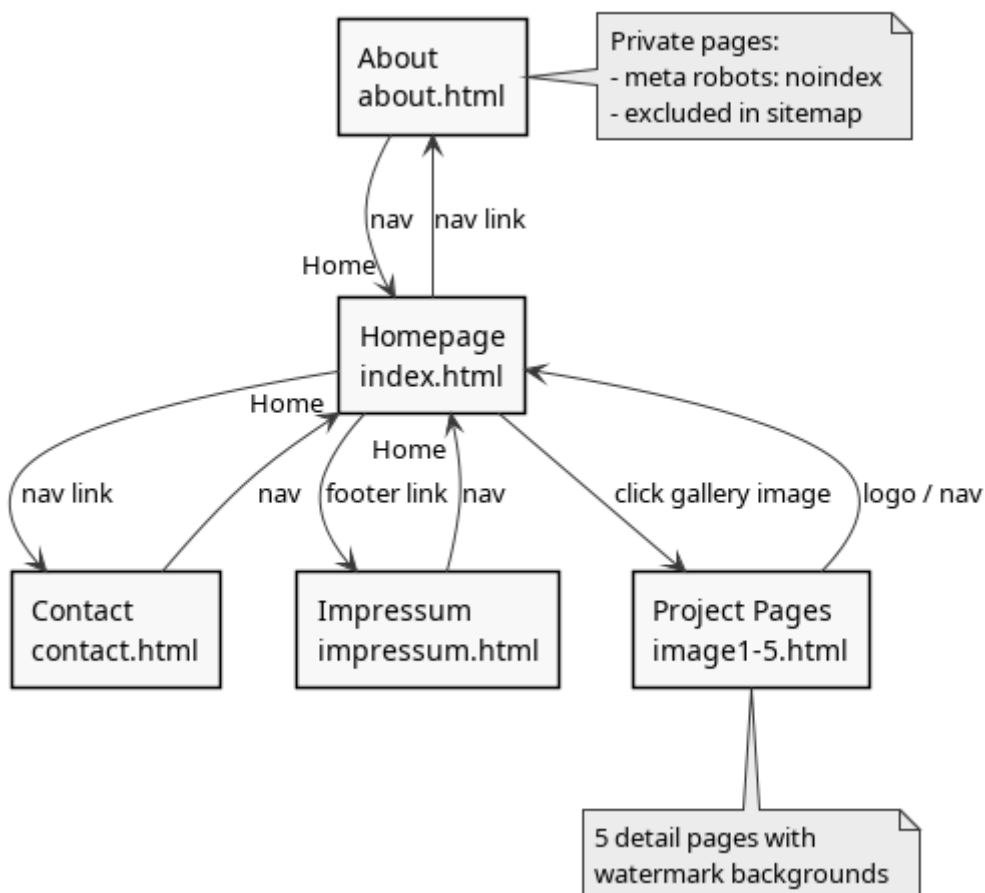
webup/
├── .git/                # Git repository
├── .gitignore           # Git ignore patterns
├── .venv/              # Python virtual environment
├── LICENSE              # MIT License
├── README.md            # Project readme
├── CHANGELOG.md         # Version history
├── CONTRIBUTING.md      # Contribution guidelines
├── docs/               # Documentation folder
│   └── index.adoc       # This documentation
└── html/               # Website root
    ├── index.html       # Homepage
    ├── about.html       # About page (noindex)
    ├── contact.html     # Contact page (noindex)
    ├── impressum.html   # Legal notice (noindex)
    ├── image1.html      # Project detail page 1
    ├── image2.html      # Project detail page 2
    ├── image3.html      # Project detail page 3
    ├── image4.html      # Project detail page 4
    ├── image5.html      # Project detail page 5
    ├── robots.txt       # Search engine directives
    ├── sitemap.xml      # SEO sitemap
    ├── humans.txt       # Credits and info
    └── css/
        └── style.css    # Main stylesheet
  
```

Central design tokens in **:root** flow into each style section enabling rapid theming and consistent spacing, color, and effect reuse.

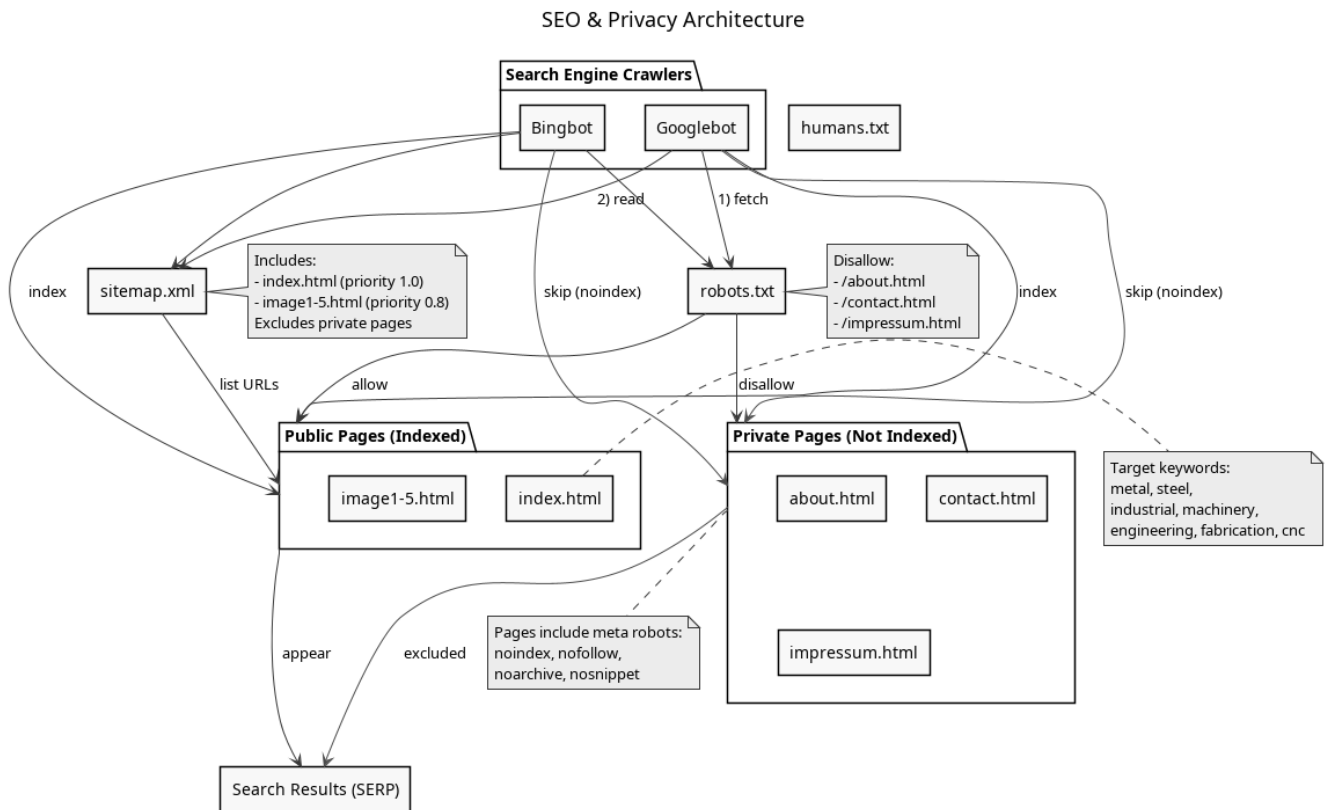
4.4. Navigation Flow

WebUp Navigation Flow (Simplified)



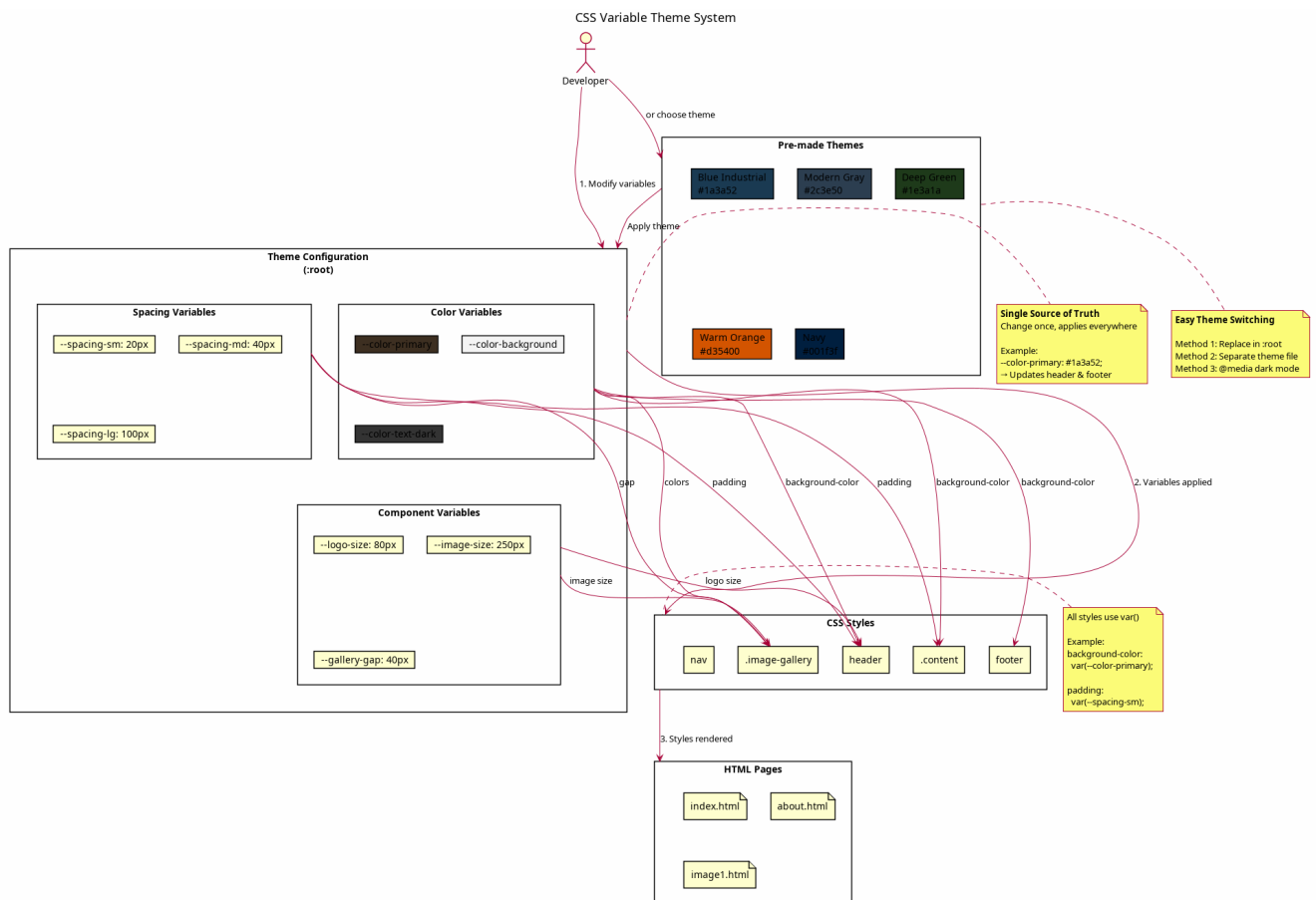
Public gallery drives traffic into project detail pages; private (noindex) informational pages and impressum remain excluded from search but reachable via global nav/footer.

4.5. SEO & Privacy Architecture



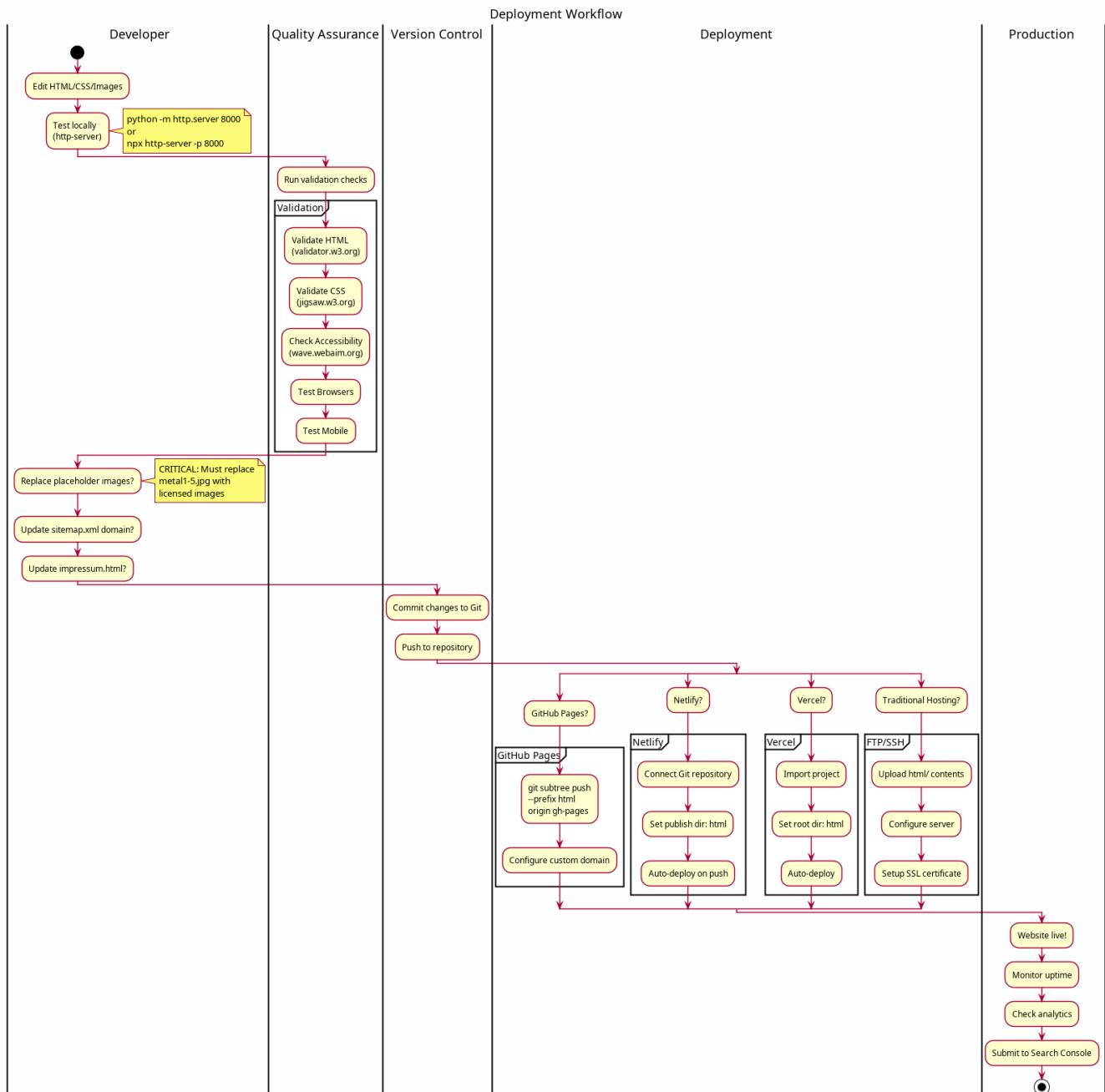
`robots.txt` and meta robots directives gate crawlers; `sitemap.xml` lists only public pages ensuring controlled search surface.

4.6. Theme System Overview



Core tokens allow swapping palettes (industrial sepia → alternative themes) without structural CSS changes.

4.7. Deployment Workflow



Local authoring → version control → optional CI → static host (GitHub Pages/Netlify/Vercel) → CDN delivery.



Looking for all diagrams on one page? See [Architecture Diagrams Gallery](#).

4.8. Section Listing (CSS Layers)

The stylesheet is segmented for maintainability:

- CSS Variables (design tokens)
- Global Styles (reset + body)
- Visually Hidden Utility (.sr-only)
- Top Navigation Bar (fixed position with logo and menu)
- Header Styles (centered title)

- Content Area Styles
- Watermark Background Styles
- Image Gallery Styles
- Footer Styles (scrolling with validator badges)
- Responsive Media Queries (tablet and mobile breakpoints)

Each segment consumes tokens from `:root` for consistent theming.

5. Navigation System

5.1. Fixed Top Navigation Bar

The website features a modern fixed navigation bar that stays visible while scrolling:

- **Position:** Fixed at the top of the viewport (z-index: 1000)
- **Logo:** 32px gear logo on the left, links to homepage
- **Menu Items:** Centered button-style links (Home, About, Contact, Projects)
- **Dropdown:** Projects menu reveals links to all 5 project detail pages
- **Active State:** Current page highlighted with background and border
- **Responsive:** Scales down for tablet (28px logo) and mobile (24px logo)

```
<nav class="top-nav" aria-label="Main navigation">
  <div class="logo">
    <a href="index.html" aria-label="WebUp Home">
      
    </a>
  </div>
  <ul>
    <li><a href="index.html" class="nav-button" aria-current="page">Home</a></li>
    <li><a href="about.html" class="nav-button">About</a></li>
    <li><a href="contact.html" class="nav-button">Contact</a></li>
    <li class="dropdown">
      <button class="nav-button dropdown-toggle" aria-haspopup="true" aria-
expanded="false">
        Projects ▾
      </button>
      <ul class="dropdown-menu">
        <li><a href="image1.html">Project 1</a></li>
        <li><a href="image2.html">Project 2</a></li>
        <li><a href="image3.html">Project 3</a></li>
        <li><a href="image4.html">Project 4</a></li>
        <li><a href="image5.html">Project 5</a></li>
      </ul>
    </li>
  </ul>
</nav>
```

5.2. Button-Style Menu Items

Navigation items are styled as modern buttons with:

- Transparent background with subtle borders
- Padding: 8px vertical, 16px horizontal
- Border radius for rounded corners
- Hover effect: semi-transparent white background
- Focus outline for keyboard navigation
- Active page indicator: darker background and visible border

```
.nav-button {
  color: var(--color-white);
  background: transparent;
  border: 1px solid transparent;
  border-radius: 4px;
  padding: 8px 16px;
  transition: all var(--transition-fast);
}

.nav-button:hover {
  background-color: rgba(255, 255, 255, 0.1);
  border-color: rgba(255, 255, 255, 0.3);
}
```

5.3. Dropdown Menu

The Projects dropdown provides access to all project detail pages:

- **Hidden by Default:** `display: none`, `visibility: hidden`, `opacity: 0`
- **Activated on Hover:** Uses `:hover` and `:focus-within` pseudo-classes
- **Smooth Transition:** Fades in with opacity transition
- **Elevated:** Box shadow creates depth and separation
- **Accessible:** `aria-haspopup` and keyboard navigation support

```
.dropdown-menu {
  display: none;
  visibility: hidden;
  opacity: 0;
  position: absolute;
  top: 100%;
```

```

    left: 0;
    background-color: var(--color-white);
    min-width: 160px;
    box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
    transition: opacity var(--transition-fast);
}

.dropdown:hover .dropdown-menu,
.dropdown:focus-within .dropdown-menu {
    display: block;
    visibility: visible;
    opacity: 1;
}

```

6. Features in Detail

6.1. Responsive Design

The website adapts to different screen sizes using:

- **CSS Media Queries:** Two breakpoints at 768px (tablet) and 480px (mobile)
- **Flexbox Layouts:** Flexible navigation and content arrangement
- **Flexible Images:** `max-width: 100%` and explicit sizing for breakpoints
- **Viewport Meta Tag:** Ensures proper mobile rendering
- **Responsive Navigation:** Logo and menu scale down on smaller screens
- **Responsive Typography:** Font sizes reduce at tablet and mobile breakpoints
- **Responsive Images:** Gallery boxes scale from 250px → 180px → 140px
- **Single Column Layout:** Gallery switches to 1 column on mobile

6.1.1. Breakpoint Details

Element	Desktop (>768px)	Tablet (≤768px)	Mobile (≤480px)
Navigation Logo	32px	28px	24px
Header Margin	48px	40px	32px
Gallery Images	250px	180px	140px
Image Padding	15px	10px	8px
Gallery Gap	40px	30px	20px
H1 Font Size	Default	1.5rem	1.2rem

Element	Desktop (>768px)	Tablet (≤768px)	Mobile (≤480px)
Body Font Size	Default	0.95rem	0.85rem
Nav Button Padding	8px 16px	6px 12px	5px 8px

6.2. Accessibility

WCAG compliance features:

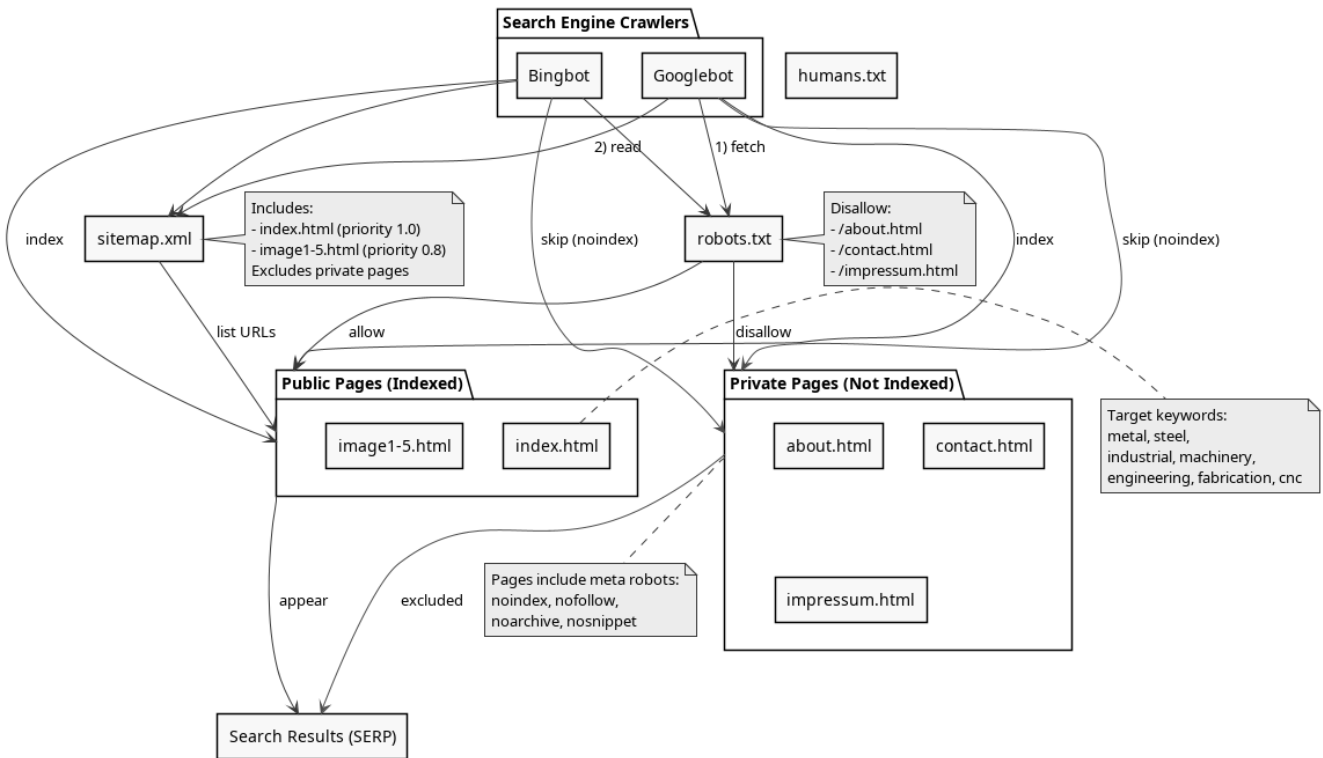
- Semantic HTML5 elements (`<header>`, `<nav>`, `<main>`, `<footer>`, `<figure>`)
- ARIA labels for navigation (`aria-label`, `aria-current`)
- ARIA attributes for dropdown (`aria-haspopup`, `aria-expanded`)
- Proper heading hierarchy (h1 → h2)
- Descriptive alt text for images
- Keyboard navigation support with focus styles
- Focus management with `:focus-within` for dropdowns
- Screen-reader-only content with `.sr-only` class
- Sufficient color contrast ratios
- Skip-to-content functionality (implicit with fixed nav)
- Sufficient color contrast ratios

6.3. SEO Optimization

Search engine optimization includes:

- Meta descriptions on all pages
- Keyword targeting: metal, steel, industrial, machinery, engineering, fabrication, cnc
- Semantic HTML structure
- `robots.txt` for crawler control
- `sitemap.xml` for search engines
- Proper page titles
- Author attribution

SEO & Privacy Architecture



6.4. Privacy Compliance

Privacy-focused features:

- Three pages excluded from search engines (about, contact, impressum)
- `noindex`, `nofollow`, `noarchive`, `nosnippet` meta tags
- robots.txt disallow directives
- Legal notice page (impressum)

6.5. CSS Variables Theming

Easy theme customization through CSS variables:

```
:root {  
  /* Colors */  
  --color-primary: #3d2f21;  
  --color-background: #f4f4f4;  
  
  /* Spacing */  
  --spacing-sm: 20px;  
  --spacing-md: 40px;  
  
  /* Logo */  
  --logo-size: 80px;  
  
  /* Gallery */
```

```
--image-size: 250px;
--image-rotation-large: 6deg;
}
```

See [Theme Variations Guide](#) for alternative color schemes.

7. Customization Guide

7.1. Changing Colors

Edit the CSS variables in `css/style.css`:

```
:root {
  --color-primary: #1a3a52;      /* New header/footer color */
  --color-background: #f0f4f8;   /* New background color */
}
```

7.2. Changing Logo

1. Replace `images/gear.png` with your logo (recommended: 80x80px)
2. Replace `images/gear.ico` with your favicon
3. Update the CSS variable if using different size:

```
:root {
  --logo-size: 100px; /* Adjust as needed */
}
```

7.3. Adding New Pages

1. Create new HTML file in `html/` directory
2. Copy structure from existing page
3. Update navigation links in all files
4. Add to `sitemap.xml` if publicly accessible
5. Update `robots.txt` if page should be excluded

7.4. Modifying Gallery

Change image count or arrangement in `index.html`:

```
<section class="image-gallery" aria-label="Project Gallery">
  <article class="image-container rotate-left">
```

```
<a href="image1.html">
  
</a>
</article>
<!-- Add more images -->
</section>
```

Rotation classes available:

- `rotate-left` (-6deg)
- `rotate-left-center` (-3deg)
- `rotate-center` (0deg)
- `rotate-right-center` (+3deg)
- `rotate-right` (+6deg)

8. Deployment

8.1. Static Hosting

Deploy to any static hosting service:

8.1.1. GitHub Pages

```
# Push html/ folder to gh-pages branch
git subtree push --prefix html origin gh-pages
```

8.1.2. Netlify

1. Connect your GitHub repository
2. Set build directory to `html`
3. Deploy

8.1.3. Vercel

1. Import your GitHub repository
2. Set output directory to `html`
3. Deploy

8.2. Apache Configuration

If using Apache, add `.htaccess` for security headers:

```
# Security Headers
Header set X-Frame-Options "SAMEORIGIN"
Header set X-Content-Type-Options "nosniff"
Header set X-XSS-Protection "1; mode=block"
Header set Referrer-Policy "strict-origin-when-cross-origin"

# Enable compression
<IfModule mod_deflate.c>
    AddOutputFilterByType DEFLATE text/html text/css text/javascript
</IfModule>

# Browser caching
<IfModule mod_expires.c>
    ExpiresActive On
    ExpiresByType image/jpeg "access plus 1 year"
    ExpiresByType image/png "access plus 1 year"
    ExpiresByType text/css "access plus 1 month"
</IfModule>
```

8.3. Pre-Deployment Checklist

Before deploying to production:



1. ☐ Replace placeholder images with licensed images
2. ☐ Update impressum.html with actual company information
3. ☐ Change domain in sitemap.xml
4. ☐ Test all links and navigation
5. ☐ Validate HTML (<https://validator.w3.org/>)
6. ☐ Validate CSS (<https://jigsaw.w3.org/css-validator/>)
7. ☐ Test accessibility (<https://wave.webaim.org/>)
8. ☐ Test on multiple browsers and devices
9. ☐ Check mobile responsiveness
10. ☐ Verify robots.txt configuration

9. Maintenance

9.1. Regular Updates

- Update project images and descriptions
- Refresh meta descriptions for SEO
- Check for broken links

- Update copyright year in footer
- Review and update CHANGELOG.md

9.2. Browser Testing

Test regularly in:

- Google Chrome (latest)
- Mozilla Firefox (latest)
- Apple Safari (latest)
- Microsoft Edge (latest)
- Mobile browsers (iOS Safari, Chrome Mobile)

9.3. Performance Optimization

1. Optimize images (compress, use WebP)
2. Minify CSS for production
3. Enable server compression (gzip)
4. Implement browser caching
5. Use a CDN for static assets

10. Troubleshooting

10.1. Common Issues

10.1.1. Images Not Loading

Problem: Images don't display on live site

Solution: Check file paths are relative and case-sensitive

```
<!-- Correct -->


<!-- Incorrect -->


```

10.1.2. Fixed Footer Overlapping Content

Problem: Footer covers bottom content

Solution: Content has `padding-bottom: 100px` to prevent overlap

10.1.3. Logo Position Wrong

Problem: Logo doesn't align properly

Solution: Adjust CSS variables:

```
:root {  
  --logo-position-left: 20px; /* Adjust horizontal position */  
  --logo-size: 80px;          /* Adjust size */  
}
```

10.1.4. Gallery Not Responsive

Problem: Images overflow on mobile

Solution: Gallery uses `flex-wrap: wrap` by default. Check CSS variables:

```
:root {  
  --image-size: 250px; /* Reduce for smaller screens */  
  --gallery-gap: 40px; /* Reduce spacing */  
}
```

11. Contributing

See [CONTRIBUTING.md](#) for contribution guidelines.

11.1. Code Style

- Use 4 spaces for indentation
- Add comments for complex sections
- Follow semantic HTML conventions
- Maintain accessibility features
- Test across browsers

12. License

This project is licensed under the MIT License. See [LICENSE](#) file for details.



Image Licensing Notice:

The placeholder images (metal1-5.jpg, gear.png, gear.ico) are NOT covered by the MIT License and must be replaced with properly licensed images before

13. Support

- GitHub Issues: <https://github.com/iosan/webup/issues>
- Repository: <https://github.com/iosan/webup>
- Branch: v0.5.1

14. Appendix

14.1. File Reference

14.1.1. index.html

Homepage with 5-image project gallery

14.1.2. about.html

Company information page (excluded from search engines)

14.1.3. contact.html

Contact information page (excluded from search engines)

14.1.4. impressum.html

Legal notice required in Germany (excluded from search engines)

14.1.5. image1-5.html

Individual project detail pages with watermarked backgrounds

14.1.6. robots.txt

Directives for web crawlers

14.1.7. sitemap.xml

XML sitemap for search engines

14.1.8. humans.txt

Credits and technology information

14.2. Keywords

The website targets these SEO keywords:

- metal
- steel
- industrial
- machinery
- engineering
- fabrication
- cnc

14.3. Browser Support

Browser	Minimum Version
Chrome	Latest
Firefox	Latest
Safari	Latest
Edge	Latest
Opera	Latest

Last Updated: November 6, 2025

Version: 0.5.1

Author: iosan